

1. Use the equation of the polynomial function $f(x) = 4x^3 - x$ to complete the following.

- (a) State the degree and the leading coefficient.
 (b) Describe the end behavior of the graph of the function.
 (c) Support your answer by graphing the function.

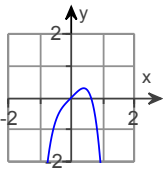
(a) The degree of the polynomial is and the leading coefficient is .

(b) Describe the end behavior of the graph of the function.

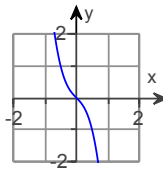
The curve opens (1) _____ to the right because the leading coefficient is (2) _____. Because the polynomial is (3) _____ the graph has end behaviors in the (4) _____ direction, so the other end opens (5) _____ to the left.

(c) Which of the following is the correct graph of the function $f(x) = 4x^3 - x$?

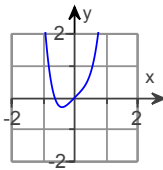
☐ A.



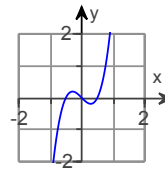
☐ B.



☐ C.



☐ D.



- (1) ☐ up (2) ☐ positive (3) ☐ quartic, (4) ☐ same (5) ☐ down
☐ down ☐ negative ☐ cubic, ☐ opposite ☐ up

2. The profit function for a product is given by $P(x) = -x^3 + 3x^2 + 400x - 500$, where x is the number of units produced and sold and P is in hundreds of dollars. Find the numbers of units that will give a profit of \$70,000.

The numbers of units that will give a profit of \$70,000 are .

(Type an integer or a fraction. Use a comma to separate answers as needed.)

3. The weekly revenue for a product is given by $R(x) = 302.4x - 0.0585x^2$, and the weekly cost is $C(x) = 10,000 + 151.2x - 0.117x^2 + 0.00003x^3$, where x is the number of units produced and sold.

- (a) How many units will give the maximum profit?
 (b) What is the maximum possible profit?

(a) The number of units that will give the maximum profit is .

(Round to the nearest whole number as needed.)

(b) The maximum possible profit is \$.

4.

Full data set

The adjacent table gives the number of worldwide Internet users, in millions.

Complete parts (a) through (b).

Year	Users (millions)	Year	Users (millions)
1995	15	2002	590
1996	32	2003	715
1997	73	2004	829
1998	181	2005	1056
1999	287	2006	1093
2000	339	2007	1232
2001	534		

(a) Find the cubic function that is the best fit for the data, with y equal to the number of millions of users and x equal to the number of years from 1990.

$y =$

(Use integers or decimals for any numbers in the expression. Round to three decimal places as needed.)

(b) Use the model to predict the number of users in 2016.

The number of users in 2016 will be approximately million.

(Round to the nearest whole number as needed.)

5. The size of the civilian labor force (in millions) for selected years from 1950 and projected to 2050 can be modeled by

$y = -0.0001x^3 + 0.0058x^2 + 1.43x + 56.7$, where x is the number of years after 1950.

(a) What does this model project the civilian labor force to be in 2013?

(b) In what year does the model project the civilian labor force to be 153 million?

(a) In 2013 the model projects the civilian labor force to be million.

(Type an integer or decimal rounded to one decimal place as needed.)

(b) The model projects the civilian labor force to be 153 million in

(Round up to the next year.)